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$$\begin{aligned}& \int e^x \cos(x) dx \\& \begin{cases} u = \cos(x) \\ dv = e^x dx \end{cases} \Rightarrow \begin{cases} du = -\sin(x) dx \\ v = e^x \end{cases} \\& \int e^x \cos(x) dx = e^x \cos(x) - \int e^x (-\sin(x)) dx \\& = e^x \cos(x) + \int e^x \sin(x) dx \\& J = \int e^x \sin(x) dx \\& \begin{cases} u = \sin(x) \\ dv = e^x dx \end{cases} \Rightarrow \begin{cases} du = \cos(x) dx \\ v = e^x \end{cases} \\& = e^x \sin(x) - \int e^x \cos(x) dx \\& J = e^x \sin(x) - I \\& I = e^x \cos(x) + J \\& I = e^x \cos(x) + e^x \sin(x) - I \\& 2I = e^x \cos(x) + e^x \sin(x) + C \\& I = \frac{e^x}{2} (\cos(x) + \sin(x)) + C\end{aligned}$$

References